

Question	Marking Guidance	Mark	Comments
05.1	1. Joins (adjacent DNA) nucleotides; 2. (Catalyses) condensation (reactions); 3. (Catalyses formation of) phosphodiester bonds (between adjacent nucleotides);	2 max	1. Reject suggestions that it forms hydrogen bonds or joins complementary bases. 1. Reject 'nucleotide bases'.
05.2	Final answer with 2sf or 3sf in range 31.8 to 34.7%;; 1 mark for 5.5 to 6.1 hours OR Final answer with 2sf or 3sf in range 46.6 to 53.0% OR Correct final answers rounded to more than 3sf OR Final answer with 2sf or 3sf in range 30.8 to 31.7 or 34.8 to 35.6%.	2	
05.3	1. Attachment/association of (inorganic) phosphate (to the enzyme); 2. (Released from) hydrolysis of ATP OR (Released from) ATP to ADP + Pi;	2	1 and 2. For 'phosphate/Pi' accept PO_4^{3-} and P in a circle. 1. Accept 'phosphate goes to the enzyme'. 1. Ignore named bonding or position of phosphate attaching to enzyme but reject formation of E-S complex. 2. For ATP accept adenosine triphosphate. 2. For ADP accept adenosine diphosphate.

05.4	1. Shortens interphase OR Cells begin DNA replication earlier OR DNA replication (starts) faster; 2. Fast(er) cell cycle/division/multiplication/mitosis OR Uncontrolled cell division/mitosis; 3. (Resulting in) a mass/group of abnormal/excessive cells;	2 max	1. Accept 'starts mitosis earlier'. 2. Accept '(May result in) mutation in a tumour suppressor gene' OR '(May result in) mutation in an oncogene'. 2 and 3. Ignore uncontrolled growth; 3. Accept '(Resulting in) a growth of abnormal/excessive cells'.
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